A ZARISKI REFINEMENT OF A FAMILY OF MORPHISMS GIVES AN EQUIVALENCE OF CATEGORIES OF DESCENT DATA

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's parenthetical '(details omitted)' attached to the cocycle condition for the glued transition morphisms; the glueing itself is displayed and the omitted verification is routine. The parenthetical is reproduced verbatim and the verification is recorded as an obligation of the card.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Morphisms of families of maps with fixed target and descent data relative to a family. Sites, Definition sites-definition-morphism-coverings, invoked at proof lines 9025-9028, together with the same chapter's Lemma lemma-pullback-family invoked at proof line 9040. Fixes the functor whose equivalence is asserted.
- Zariski coverings of a scheme. Topologies, Definition topologies-definition-zariski-covering, invoked at proof lines 9030-9031. Supplies the covering property that makes the pulled back transition isomorphisms a descent datum amenable to glueing.
- Full faithfulness of the pullback functor along an fpqc refinement. Same chapter, Lemma lemma-fpqc-refinement-coverings-fully-faithful, invoked at proof lines 9047-9048. Gives faithfulness and full faithfulness of the functor, reducing the lemma to essential surjectivity.
- Glueing of schemes along open subschemes. Schemes, Section schemes-section-glueing-schemes, invoked at proof lines 9063-9065. Produces the scheme over V_j from the descent datum over the Zariski covering of V_j.

Source reference. Original lemma 02VZ: sources/stacks-descent/pinned-original-source.tex, source lines 8952–8952. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02VS: sources/stacks-descent/pinned-original-source.tex, source lines 8570–8570. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 01JA: sources/stacks-chow/tags.txt, source lines 2007–2007. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 00VT: sources/stacks-chow/tags.txt, source lines 1159–1159. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 020O: sources/stacks-chow/tags.txt, source lines 2641–2641. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{U} = \{U_i \to S\}_{i\in I}$, and $\mathcal{V} = \{V_j \to S\}_{j\in J}$, be families of morphisms with target S. Let $\alpha : I \to J$, $id : S \to S$ and $g_i : U_i \to V_{\alpha(i)}$ be a morphism of families of maps with fixed target, see Sites, Definition 00VT. Assume that for each $j \in J$ the family $\{g_i : U_i \to V_j\}_{\alpha(i)=j}$ is a Zariski covering (see Topologies, Definition 020O) of V_j. Then the pullback functor*

$$\textit{descent data relative to } \mathcal{V} \longrightarrow \textit{descent data relative to } \mathcal{U}$$

of Lemma 02VS is an equivalence of categories. In particular, the category of schemes over S is equivalent to the category of descent data relative to any Zariski covering of S.

Proof. The functor is faithful and fully faithful by Lemma 02VZ. Let us indicate how to prove that it is essentially surjective. Let $(X_i, \varphi_{ii'})$ be a descent datum relative to $\mathcal{U}$. Fix $j \in J$ and set $I_j = \{i \in I \mid \alpha(i) = j\}$. For $i, i' \in I_j$ note that there is a canonical morphism

$$c_{ii'} : U_i \times_{g_i, V_j, g_{i'}} U_{i'} \to U_i \times_S U_{i'}.$$

Hence we can pullback $\varphi_{ii'}$ by this morphism and set $\psi_{ii'} = c_{ii'}^* \varphi_{ii'}$ for $i, i' \in I_j$. In this way we obtain a descent datum $(X_i, \psi_{ii'})$ relative to the Zariski covering $\{g_i : U_i \to V_j\}_{i\in I_j}$. Note that $\psi_{ii'}$ is an isomorphism from the open $X_{i, U_i \times_{V_j} U_{i'}}$ of X_i to the corresponding open of $X_{i'}$. It follows from Schemes, Section 01JA that we may glue $(X_i, \psi_{ii'})$ into a scheme Y_j over V_j. Moreover, the morphisms $\varphi_{ii'}$ for $i \in I_j$ and $i' \in I_{j'}$ glue to a morphism $\varphi_{jj'} : Y_j \times_S V_{j'} \to V_j \times_S Y_{j'}$ satisfying the cocycle condition (details omitted). Hence we obtain the desired descent datum $(Y_j, \varphi_{jj'})$ relative to $\mathcal{V}$. □